Problem Set 4: optimal bundles and kinked budget lines

Microeconomics (040184), Group 1, WS 2026/27

Deadline: Wednesday 11 November 2026, 16:45

How this problem set works

Problem 1 is optional and is not part of the quiz. It is an opportunity to work through the solution with an LLM and learn it.

Problem 2 should first be attempted without an LLM, in order to test one's own understanding. It is a different problem, but a student who has understood Problem 1 should be able to solve it. An LLM may be consulted in case of difficulty, but only after one's own attempt.

Only Problem 2 is in the quiz.

Optimization with a single budget line should become very easy. Students should then become proficient with questions with a kinked budget line.

Next week we will cover problems of this difficulty with a different utility function, so it is important to be able to solve problems at this level independently.

Problem 3 concerns the translation of verbal descriptions into equations.

Answers are entered in the Moodle quiz *PS4 (Group 1)*, which opens by Monday 9 November at the latest. Only the Moodle answers are graded, and each student submits individually. The deadline is **Wednesday 11 November 2026, 16:45**.

Draw a graph for every part.

Problem 1: kinked budget lines

Part A: A single budget line

Given

A consumer has the utility function $u = 3xy$. The price of each good is €1 and the consumer's income is €10, so the budget line is $x + y = 10$. Consider the bundles P $= (2, 8)$ and Q $= (1, 9)$.

(a) Draw the budget line in a graph, with x on the horizontal axis, and mark the bundles P and Q.

(b) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle P $= (2, 8)$.

(c) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle Q $= (1, 9)$.

(d) The MRS differs between the bundles P and Q. Which of the following statements explain why?

I. MU_x depends on y.
II. MU_y depends on x.
III. The prices differ between P and Q.

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

(e) Which of the following statements correctly interpret $\frac{\text{MU}_x}{p_x}$?

I. The "bang for the buck" on x.
II. The marginal utility of x per euro spent on x.
III. The marginal utility of x, normalized by the price of y.

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

(f) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part B: A change in income

Given

The utility function is still $u = 3xy$ and both prices are still €1, but the consumer's income rises from €10 to €18, so the budget line is now $x + y = 18$. Consider the bundles P $= (2, 16)$ and Q $= (3, 15)$.

(a) Draw the new budget line in the same graph as the old one, and mark the bundles P and Q.

(b) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle P $= (2, 16)$.

(c) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle Q $= (3, 15)$.

(d) Compare the bundle $(2, 16)$ with the bundle $(2, 8)$ from Part A. The bundle $(2, 16)$ contains the same quantity of x and a larger quantity of y. Which of the following statements is correct?

- ☐ MU_x is higher, MU_y is the same, and the MRS is higher
- ☐ MU_x is lower, MU_y is the same, and the MRS is lower
- ☐ MU_x is the same, MU_y is higher, and the MRS is lower
- ☐ both marginal utilities are higher, and the MRS is the same
- ☐ both marginal utilities are the same, and the MRS is the same

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part C: A change in price

Given

The utility function is still $u = 3xy$ and the income is still €18, but the price of x rises from €1 to €3, so the budget line is now $3x + y = 18$. Consider the bundles $\text{P} = (1, 15)$ and $\text{Q} = (2, 12)$.

(a) Draw the new budget line in the same graph, and mark the bundles P and Q.

(b) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle $\text{P} = (1, 15)$.

(c) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle $\text{Q} = (2, 12)$.

(d) The price of x has changed since Part B. At a given bundle, which of the following changes as a result?

- ☐ MU_x and the MRS
- ☐ MU_x, MU_y and the MRS
- ☐ only MU_x
- ☐ only $\frac{\text{MU}_x}{p_x}$
- ☐ only the MRS

(e) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part D: A kinked budget line

Given

The utility function is still $u = 3xy$. The consumer's income is €10, and the price of y is €1. The first 4 units of x cost €1 each, and each further unit costs €3. The budget line therefore has two segments: $x + y = 10$ for $x \leq 4$, and $3x + y = 18$ for $x \geq 4$.

(a) Draw the budget line in a graph, and shade the bundles the consumer can afford.

(b) What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Part E: Two further cases

Given

The utility function is still $u = 3xy$, and the price of y is €1. Two further kinked budget lines are considered.

(a) For each case, draw the budget line in a graph.

(b) Case 1: The income is €10. The first 7 units of x cost €1 each, and each further unit costs €3. The budget line is therefore $x + y = 10$ for $x \leq 7$, and $3x + y = 24$ for $x \geq 7$. What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

(c) Case 2: The income is €16. The first unit of x costs €1, and each further unit costs €3. The budget line is therefore $x + y = 16$ for $x \leq 1$, and $3x + y = 18$ for $x \geq 1$. What is the optimal bundle (x^*, y^*), the bundle that maximizes utility and satisfies the budget constraint by spending all income?

Before Problem 2: derivatives of powers and roots

The power rule. For a power α of a variable, differentiation multiplies by α and lowers the power by 1.

$$\frac{d}{dx}\, x^{\alpha} = \alpha\, x^{\alpha-1}$$

Sums. The derivative of a sum is the sum of the derivatives. A constant factor is retained.

Negative powers. A negative power denotes a reciprocal: $x^{-\frac{1}{2}} = \dfrac{1}{\sqrt{x}}$.

A square root is a power. $\sqrt{x} = x^{\frac{1}{2}}$. So its derivative is $\frac{1}{2}\, x^{-\frac{1}{2}} = \dfrac{1}{2\sqrt{x}}$.

A constant inside the root. $\sqrt{9x} = 3\sqrt{x}$, so its derivative is $3 \cdot \dfrac{1}{2\sqrt{x}} = \dfrac{3}{2\sqrt{x}}$.

Two goods. When differentiating with respect to x, y is treated as a constant. $10\sqrt{xy} = 10\sqrt{y}\cdot\sqrt{x}$, so

$$\mathrm{MU}_x = 10\sqrt{y} \cdot \frac{1}{2\sqrt{x}} = 5\sqrt{\frac{y}{x}}$$

Similarly, $\mathrm{MU}_y = 5\sqrt{\dfrac{x}{y}}$. Problem 2 uses the coefficient 12 instead of 10.

Problem 2: kinked budget lines and a square root

Part A: A single budget line

Given

A consumer has the utility function $u = 12\sqrt{xy}$. The price of each good is €1 and the consumer's income is €24, so the budget line is $x + y = 24$. Consider the affordable bundles $\mathrm{P} = (4, 16)$ and $\mathrm{Q} = (2, 18)$.

(a) Draw the budget line in a graph, with x on the horizontal axis, and mark the bundles P and Q.

(b) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle $\mathrm{P} = (4, 16)$.

(c) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle $\mathrm{Q} = (2, 18)$.

(d) The MRS differs between the bundles P and Q. Which of the following statements explain why?

I. $\frac{y}{x}$ differs between P and Q.
II. The coefficient 12 in the utility function differs between P and Q.
III. Each marginal utility depends on both x and y.

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

(e) Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

Part B: A change in income

Given

The utility function is still $u = 12\sqrt{xy}$ and both prices are still €1, but the consumer's income rises from €24 to €48, so the budget line is now $x + y = 48$. Consider the affordable bundles P $= (8, 32)$ and Q $= (4, 36)$.

(a) Draw the new budget line in the same graph as the old one, and mark the bundles P and Q.

(b) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle P $= (8, 32)$.

(c) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle Q $= (4, 36)$.

(d) The bundle $(4, 16)$ from Part A and the bundle $(8, 32)$ have the same MRS. What is the reason?

- ☐ the ratio $\frac{y}{x}$ is the same at both bundles
- ☐ both bundles lie on a budget line
- ☐ income has risen
- ☐ the prices are the same
- ☐ the utility is the same at both bundles

(e) Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

Part C: A change in price

Given

The utility function is still $u = 12\sqrt{xy}$ and the income is still €48, but the price of x rises from €1 to €4, so the budget line is now $4x + y = 48$. Consider the affordable bundles P = (1, 36) and Q = (8, 8).

(a) Draw the new budget line in the same graph, and mark the bundles P and Q.

(b) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle P = (1, 36).

(c) Calculate the marginal utility of x, the marginal utility of y and the marginal rate of substitution at the bundle Q = (8, 8).

(d) The price of x has changed since Part B. At a given bundle, which of the following changes as a result?

- ☐ MU_x and the MRS
- ☐ MU_x, MU_y and the MRS
- ☐ only MU_x
- ☐ only $\frac{\text{MU}_x}{p_x}$
- ☐ only the MRS

(e) Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

Part D: Two budgets available

Given

The utility function is still $u = 12\sqrt{xy}$. Now both budgets are available to the consumer, who can choose along either one: $4x + y = 48$ from Part C and $x + y = 24$ from Part A. The two lines intersect, and the outer boundary has an inward kink there, closer to the origin than the two ends of the boundary. To the left of the kink, $4x + y = 48$ is the outer line; to the right of the kink, $x + y = 24$ is the outer line. The outer boundary is therefore $4x + y = 48$ for $x \leq 8$, and $x + y = 24$ for $x \geq 8$.

(a) Draw both budget lines in a graph, and shade the bundles the consumer can afford.

(b) Which of the following gives the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income?

- ☐ only $(8, 16)$
- ☐ both $(4, 32)$ and $(16, 8)$
- ☐ only $(12, 12)$
- ☐ both $(6, 24)$ and $(12, 12)$
- ☐ only $(6, 24)$

Part E: Two further cases

Given

The utility function is still $u = 12\sqrt{xy}$. In each case, two budgets are again available, and the consumer can choose along either one.

(a) For each case, draw the budget line in a graph.

(b) Case 1: Both $4x + y = 48$ and $x + y = 36$ are available. The outer boundary is $4x + y = 48$ for $x \leq 4$, and $x + y = 36$ for $x \geq 4$. Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

(c) Case 2: Both $4x + y = 72$ and $x + y = 24$ are available. The outer boundary is $4x + y = 72$ for $x \leq 16$, and $x + y = 24$ for $x \geq 16$. Determine the optimal bundle or bundles (x^*, y^*), that is, the bundle or bundles that maximize utility and satisfy the budget constraint by spending all income.

Problem 3: from words to equations

Part A: Perfect substitutes

In each of the following, a consumer's preferences are described in words. Which utility function represents them?

(a) One pack of pasta makes one meal, and the consumer is indifferent between two brands of pasta. Let x denote the number of packs of the first brand and y the number of packs of the second brand. The consumer cares only about the number of meals. Which utility function represents these preferences?

- ☐ $u = 2x + y$
- ☐ $u = \min\{x, y\}$
- ☐ $u = x + 4y$
- ☐ $u = x + y$
- ☐ $u = xy$

(b) A small pack contains 2 eggs and a large pack contains 3 eggs. The consumer cares only about the total number of eggs. Let x denote the number of small packs and y the number of large packs. Which utility function represents these preferences?

- ☐ $u = 2x + 3y$
- ☐ $u = 3x + 2y$
- ☐ $u = \min\{2x, 3y\}$
- ☐ $u = x + y$
- ☐ $u = xy$

(c) One hour of running burns 600 calories and one hour of cycling burns 400 calories. The consumer cares only about the total number of calories burned. Let x denote the hours of running and y the hours of cycling. Which of the following utility functions represent these preferences?

I. $u = 600x + 400y$
II. $u = 3x + 2y$
III. $u = \min\{600x, 400y\}$

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Part B: Perfect complements

In each of the following, a consumer's preferences are described in words. Which utility function represents them?

(a) A hot dog requires one sausage and one bun. A sausage without a bun yields no utility, and likewise a bun without a sausage. Let x denote the number of sausages and y the number of buns. The consumer cares only about the number of hot dogs. Which utility function represents these preferences?

- ☐ $u = \min\{3x, y\}$
- ☐ $u = \min\{x, 3y\}$
- ☐ $u = \min\{x, y\}$
- ☐ $u = x + y$
- ☐ $u = xy$

(b) Each sandwich requires 1 slice of bread and exactly 3 slices of cheese. Let x denote the number of slices of bread and y the number of slices of cheese. The consumer cares only about the number of sandwiches. Which utility function represents these preferences?

- ☐ $u = \min\{3x, \frac{y}{3}\}$
- ☐ $u = \min\{\frac{x}{3}, y\}$
- ☐ $u = \min\{x, 3y\}$
- ☐ $u = \min\{x, \frac{y}{3}\}$
- ☐ $u = x + \frac{y}{3}$

(c) A cake requires 3 eggs and 5 spoons of sugar, always in that ratio. Let x denote the number of eggs and y the number of spoons of sugar. The consumer cares only about the number of cakes. Which of the following utility functions represent these preferences?

I. $u = \min\{\frac{x}{3}, \frac{y}{5}\}$
II. $u = \min\{5x, 3y\}$
III. $u = \min\{3x, 5y\}$

- ☐ only I
- ☐ only II
- ☐ I and II
- ☐ I and III
- ☐ I, II and III

Part C: Budget lines

In each of the following, a consumer's budget is described in words. Which equations describe the budget line?

(a) A consumer has €24. The price of a coffee is €2 and the price of a sandwich is €3. Let x denote the number of coffees and y the number of sandwiches. Which equation describes the consumer's budget line?

- ☐ $2x + 3y = 12$
- ☐ $2x + 3y = 24$
- ☐ $2x + 3y = 5$
- ☐ $3x + 2y = 24$
- ☐ $x + y = 24$

(b) A consumer has €30. The first 5 gym visits cost €4 each, and each further visit costs €2. All other goods cost €1 per unit. Let x denote the number of visits and y the quantity of all other goods. Which equations describe the consumer's budget line?

- ☐ $2x + y = 30$ for $x \leq 5$, and $4x + y = 30$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 4$, and $2x + y = 20$ for $x \geq 4$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 10$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 20$ for $x \geq 5$
- ☐ $4x + y = 30$ for $x \leq 5$, and $2x + y = 30$ for $x \geq 5$

(c) A consumer has €20. The first 3 bus rides are free, and each further ride costs €2. The consumer can take at most 10 rides. All other goods cost €1 per unit. Let x denote the number of rides and y the quantity of all other goods. Which equations describe the consumer's budget line?

- ☐ $2x + y = 20$ for $x \leq 10$
- ☐ $y = 20$ for $x \leq 3$; $2x + y = 20$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$; $2x + y = 23$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$; $2x + y = 26$ for $3 \leq x \leq 10$
- ☐ $y = 20$ for $x \leq 3$; $2x + y = 26$ for $x \geq 3$, with no limit on the number of rides